Assignment - 1

Section - A

ans 1.
- Hides complex details, shows essential features
- reduces system complexity
- make design & development easier
- Enable modularity

ans 2.
- Hardware : physical components (CPU, memory)
- Machine : Binary instructions
- Assembly : Symbolic instructions (ADD, MOV)
- High-Level : C, JAVA, Python
- Application : End-user programe

ans-3
- input unit :- Takes input
- memory unit : stores data/instructions
- ALU unit : performs calculations
- control unit : controls execution
- output unit : display result

ans 4.
ALU : Arithmetic & logic operations
CU : controls & coordinates operations
Registers : smallest and fastest storage
cache : stores frequenty used data

ans 5:

Von Neumann Model	Stored programe
• CPU, I/O, memory connected • same memory for data and instructions.	instructions stored in memory executed sequentially

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ans 6
- Single bus b/w memory & CPU
- causes delay - performance bottleneck

ans 7.
- Transistors double every 18-24 months

impact :-
- better performance
- lower cost
- smaller device.

ans 8.
IPC :-
- instructions executed per cycle
- Higher IPC - better performance.

ans 9.
Execution time = Total run time
clock rate = cycle / sec
CPI = cycle per instructions
IPC = 1/CPI

CPU Time = (instructions × CPI) / clock rate

ans 10.
Bit - smallest unit of memory (0/1)
Byte = 8 bits
Signed = + and - values
unsigned = only + values

ans 11.
Basic Operations :-
- Arithmetic Operators = +, -, /, *
- Logical Operator : AND, OR, NOT
- comparasion (>, <, =)
- Data movement (load / store)

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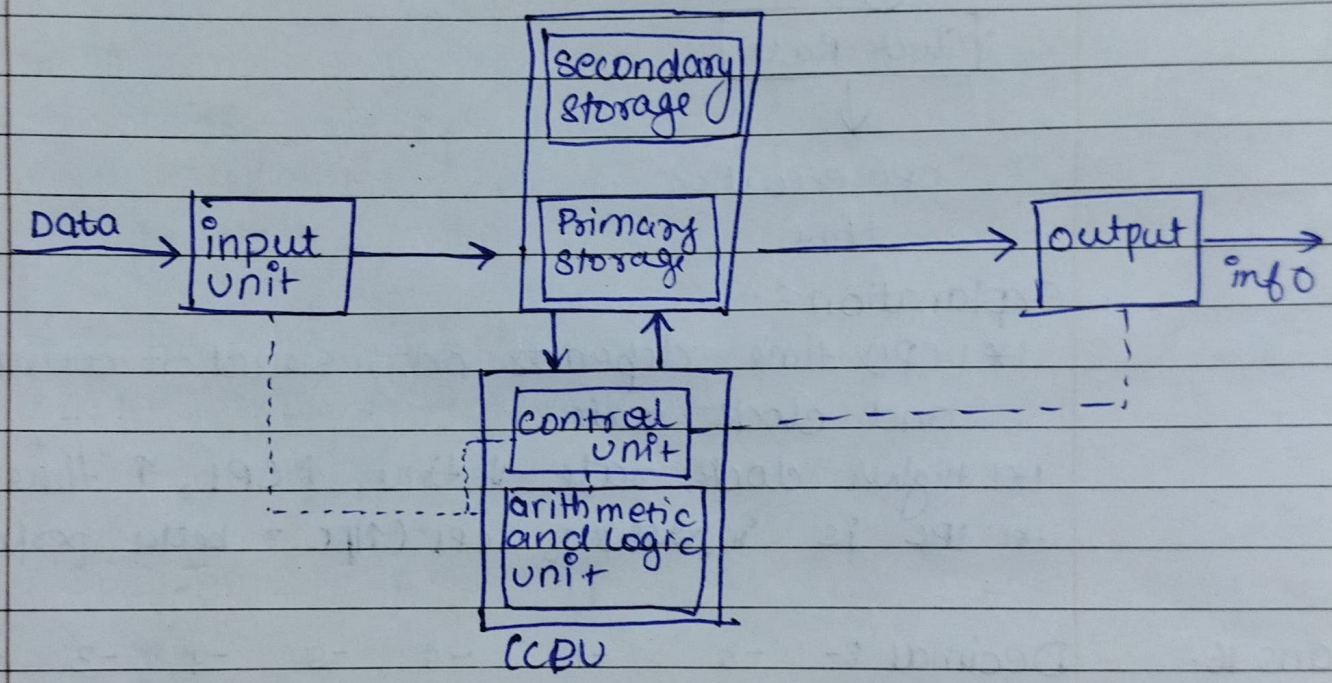
ans 12. ASCII (7/8 bit)
unicode (multi language)

Decimal
- Fixed point
- Floating point (IEEE 754)

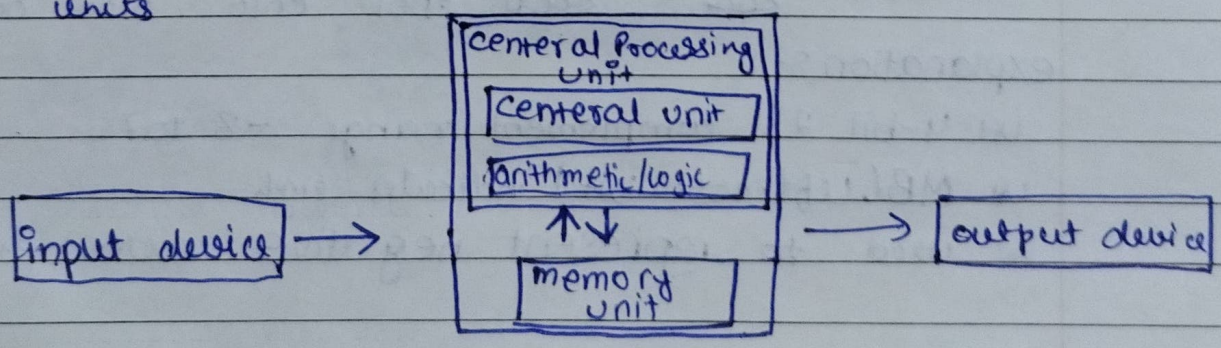
Section B

ans 13



(*) input takes data, CPU process it using ALU and CU
(*) Memory stores data and instructions
(*) output unit deploy results : buses connect all units

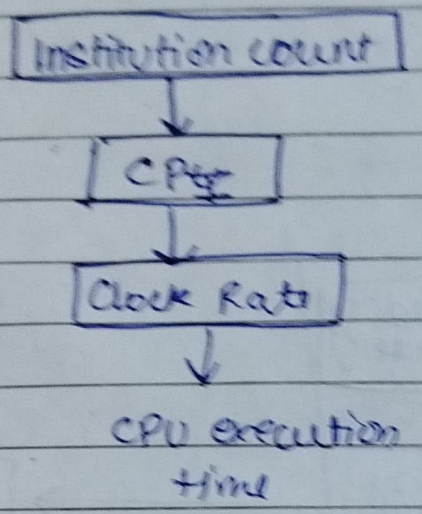
ans 14



explanation :

- stage memory stores both data and instructions
- CPU computes with memory and I/O via a common bus
- Basic of modern computers

ans 15



IPC = 1/CPI

Explanation :-

- CPU time depends on instruction count, CPI and clock rate
- Higher clock rate ↓ time, ↑ CPI, ↑ time.
- IPC is inverse of CPI (↑IPC = better performance.)

ans 16.

Decimal :-	-8	-7	-6	-5	-4	-3	-2	-1	0
Binary :-	1000	1001	1010	1011	1100	1101	1110	1111	0000

0	1	2	3	4	5	6	7
0000	0001	0010	0011	0100	0101	0110	0111

explanation :-

- 4-bit 2's complement range -8 to 7
- MB (leftmost bit) indicates sign
- used to represent negative numbers effectively

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Section C

ans 17 CPU A : 3.0 Hz CPI = 1.5
CPU B : 2.2 Hz CPI = 1.0

CPU Time = (instruction count × CPI) / clock rate

CPU A = $\frac{1.5}{3.0}$ = 0.5 CPU B = $\frac{1.0}{2.2}$ = 0.45

∴ CPU B has lower execution time, here faster
↳ Reason lower CPI outweights lower clock rate

ans 18 <u>effects of catche MPSS</u>

- catche MPSS → CPU waits (stall cycles increases)
- IPC ↓ (fewer instructions per cycle)
→ Execution time ↑

<u>fix for it</u>
- increase catche size
- improve catche locality
- use multi level cache

ans 19.
- use fixed ~~pun~~ point for given range
- faster, less memory
- simpler hardware

ans 20.
- not directly about performance
- limited by heat & power
- modern gains via parallelism
- doesnot gurantee performance doubling

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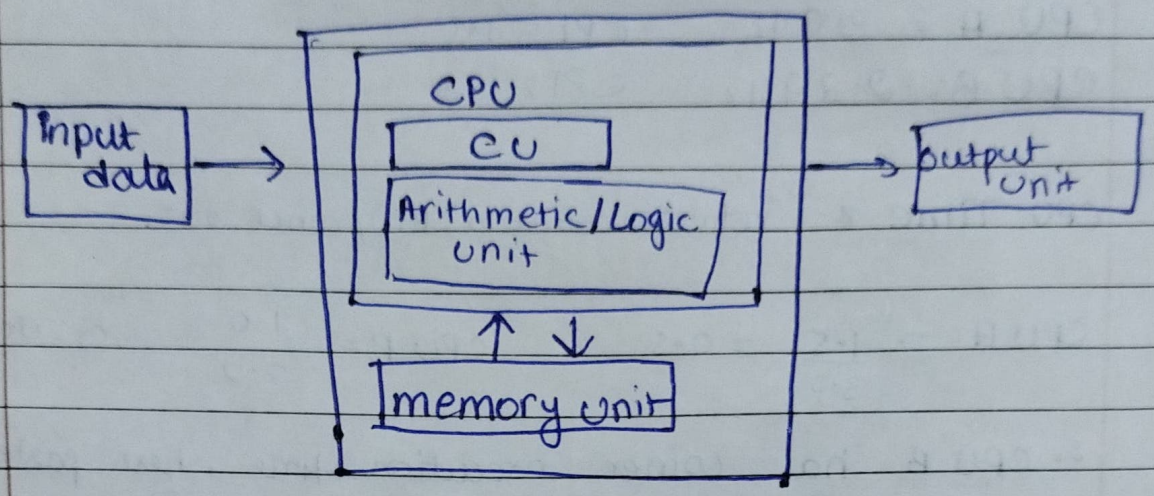
Section - D

ans 21



Von-neumon architecture

(*) design decision :- single memory for design & decision
(*) explanation :-
- simplifies system design and reduces hardware cost
- But leads to ~~nor~~ Von-Neumann bottleneck due to shared bus

ans 22 (*) Performance ratio :-

CPU Time = (instructions × CPI) / clock rate

(*) Explanation :-
- performance improves when CPU time ↓
- depends on IC, CPI, and clock rate

(*) one practical improvement
→ reduce CPI useage pipeline or better cache
→ reduce IC using optimised algo
→ increase clock rate by better hardware

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ans 23 Binary addition

+5 = 0101
+4 = 0100
result = 1001

interpretation
1001 = -7 (2's complement)

overflow
(+ve) + (+ve) → (-ve) ⇒ overflow
⇒ occurs when result exceeds range